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Understanding Your Fibromyalgia *Labs*

Which tests actually matter, what the numbers mean, and exactly what to ask your doctor for

Dr. Adam Seay, DC

Seay Wellness / Root Health, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

Contents

Please Read This First

This guide is education, not medical care. It does not diagnose you, it does not treat you, and it does not replace your physician, rheumatologist, or any other member of your care team. Reading it does not create a patient-provider relationship with Dr. Seay or Root Health.

No lab test in this guide can confirm or rule out fibromyalgia. That is not a limitation of this guide. It is a fact about the condition itself, and the rest of this document explains why that is actually useful information rather than bad news.

Never stop, start, or change the dose of any prescription medication because of something you read here. Some medications legitimately affect the lab values discussed in these pages. If that applies to you, the correct move is to tell your prescriber what you noticed, not to adjust anything yourself.

Every result in this guide has to be interpreted by a clinician who knows your full history. Numbers on a page do not carry meaning on their own. They carry meaning next to your symptoms, your exam, your medications, your family history, and your timeline. Bring your actual results to your doctor or to a Root Discovery Session.

If a result alarms you, or if you have any of the warning signs listed in the final section, get clinical review promptly rather than waiting.

The Honest Starting Point

Let us get the most important thing out of the way in the first paragraph, because most guides bury it.

There is no blood test for fibromyalgia. There is no antibody, no inflammatory marker, no hormone, no scan, and no genetic panel that confirms it. Fibromyalgia is diagnosed clinically, from your symptom pattern, using criteria built around widespread pain, fatigue, unrefreshing sleep, and cognitive symptoms, and it is confirmed by a clinician's judgment rather than by a laboratory (PMID 24737367). The 2016 revision of the diagnostic criteria is entirely symptom-based, built from a widespread pain index and a symptom severity scale, with no laboratory component at all (PMID 27916278).

You have probably been told some version of "your labs came back normal." If you took that as a verdict on whether your pain is real, please set that interpretation down. Normal labs are the expected finding in fibromyalgia. They are consistent with the diagnosis, not evidence against it.

So why run labs at all?

Because labs do three genuinely useful jobs, and none of them is "prove you have fibromyalgia."

Job one: rule out the conditions that look like fibromyalgia. Widespread pain and crushing fatigue are not unique to fibromyalgia. Thyroid disease, iron deficiency, vitamin D deficiency, B12 deficiency, celiac disease, inflammatory arthritis, and sleep apnea can all produce a picture that looks a great deal like it. Some of those are straightforward to correct. Missing one of them is the single most expensive mistake in this whole process.

Job two: find the things riding alongside it. Fibromyalgia is not exclusive. It coexists with other rheumatic diseases at meaningful rates, and someone with an inflammatory arthritis can also have fibromyalgia at the same time (PMID 24589726). One diagnosis does not close the file on the others.

Job three: describe the terrain. This is the part that most workups skip. Blood sugar regulation, inflammatory load, nutrient status, stress-axis function, and sleep quality all shape how much pain a nervous system produces and how much energy you have to spend on a given day. None of these is a fibromyalgia test. All of them are levers, and a lever you can see is a lever you can pull.

That is the honest value of testing here. Not a diagnosis. A map.

How to Read This Guide

Sections 1 and 2 cover the rule-out tier, the labs most physicians will recognize immediately and can order without any convincing. Section 3 covers the terrain tier, the labs that describe the environment your nervous system is operating in. Section 4 is about the difference between “in range” and “optimal,” which is where functional medicine and conventional medicine most often talk past each other, and where you deserve a straight answer about what is settled and what is a clinical preference. It includes the actual target windows used at Root Health. Section 5 shows you how to read your results as a pattern. Section 6 gives you the actual conversation to have with your doctor, including a request list you can print. Section 7 covers what labs cannot tell you. Section 8 covers when to stop reading and go get seen.

You do not need every test in here. Section 6 tells you where to start.

Section 1: The Rule-Out Tier, Part One (Thyroid, Iron, Vitamin D, B12)

These four account for most of the correctable conditions that get mistaken for, or that quietly worsen, a fibromyalgia picture. If you only ever get one round of labs, this is the round.

Thyroid

Underactive thyroid produces fatigue, muscle aches, cold intolerance, cognitive slowing, and low mood. Read that list again and notice how much of it overlaps with what brought you here.

The relationship goes deeper than symptom overlap. In one small controlled study, roughly a third of patients with Hashimoto's thyroiditis, the autoimmune form of thyroid disease, also met criteria for fibromyalgia, and the association tracked with the presence of thyroid antibodies rather than with subclinical hypothyroidism alone (PMID 21085966). That is a small study and it should be read as a signal rather than a settled number, but it is the reason a thorough thyroid look matters here.

What most people get ordered: TSH alone.

What is worth asking about: TSH, free T4, free T3, thyroid peroxidase antibodies (TPO), and thyroglobulin antibodies (TgAb).

Why the antibodies specifically? Because thyroid autoimmunity can be present and active while TSH still sits inside the reference range. A TSH-only panel is designed to answer "is this gland currently failing," which is a good question. It is not designed to answer "is this gland under autoimmune attack," which is a different question and the one that maps onto the association described above.

How to read it honestly. A positive TPO antibody does not mean you have fibromyalgia and it does not mean you have hypothyroidism today. It means autoimmune thyroid activity is present, which is worth knowing, worth monitoring, and worth discussing with your physician. A normal TSH with positive antibodies is a real and common pattern that deserves follow-up rather than dismissal.

Iron and Ferritin

Iron deficiency produces fatigue, exercise intolerance, poor sleep, and restless legs, all of which are already common in fibromyalgia. It can be present without anemia, which means a normal hemoglobin on a standard complete blood count does not settle the question.

The evidence here is genuinely mixed, and you should know that before you go asking for it. One case-control study found significantly lower mean ferritin in fibromyalgia patients than in matched controls, and reported that a ferritin below 50 ng/mL was associated with increased odds of fibromyalgia even when the value sat inside the laboratory's normal range (PMID 20087382). A separate study of non-anemic fibromyalgia patients found no difference in iron, ferritin, transferrin, or soluble transferrin receptor compared with controls, and concluded there was no evidence to support iron supplementation as a fibromyalgia treatment (PMID 22095117).

A randomized placebo-controlled trial of intravenous iron in patients who had both fibromyalgia and documented iron deficiency did not meet its primary endpoint, though it reported greater improvement than placebo on several secondary measures (PMID 29149437). That is a legitimately encouraging signal and it is also, honestly, a trial that missed its main target. Both things are true.

Where does that leave you? Iron status is worth checking because iron deficiency is common, correctable, and produces symptoms you already have. It is not worth treating as a fibromyalgia cure. And it is worth knowing that the exact ferritin threshold that should count as deficiency is itself an unsettled question in the literature, not a fixed line (PMID 34028001).

What to ask for: complete blood count with differential, ferritin, serum iron, total iron binding capacity, and transferrin saturation. Ask for ferritin explicitly. It is frequently left off.

One important caution: ferritin is an acute phase reactant, which means inflammation, infection, and liver issues can push it up and mask a real deficiency. That is exactly why it is read next to the other iron markers rather than alone.

Vitamin D

Vitamin D deficiency is common, easy to measure, and easy to correct, which makes it worth checking early.

A systematic review found that people with fibromyalgia tend to have lower vitamin D levels than healthy controls, while also finding conflicting results on whether supplementation improves pain or symptom control, with no clear consensus on its role in management (PMID 30886978). A small randomized placebo-controlled trial in 30 women with fibromyalgia and baseline 25-hydroxyvitamin D below 32 ng/mL supplemented to a target range of 32 to 48 ng/mL and found a significant group effect on pain scores over the treatment period, while explicitly calling for larger studies before drawing firm conclusions (PMID 24438771).

So: association is reasonably well supported, treatment benefit is promising but not established. That is the accurate summary, and anyone who tells you vitamin D fixes fibromyalgia is ahead of the evidence.

What to ask for: 25-hydroxyvitamin D, sometimes written 25(OH)D. This is the storage form and the one that reflects your status. Ask specifically for 25-hydroxy rather than 1,25-dihydroxy, which answers a different clinical question.

The number you will argue about. A 2011 Endocrine Society guideline defined vitamin D deficiency as 25(OH)D below 20 ng/mL and insufficiency as 21 to 29 ng/mL, recommending a target above 30 ng/mL for patients at risk (PMID 21646368). The Endocrine Society's 2024 guideline took a notably more conservative position, declining to recommend routine 25(OH)D testing in healthy adults and stepping back from specific target thresholds in the general population (PMID 38828931). Both are real guidelines from the same body, thirteen years apart, and they disagree. If your doctor's threshold differs from something you read online, that gap is where it comes from, and it is a reasonable thing to raise directly.

Vitamin B12 and Folate

B12 deficiency produces fatigue, cognitive fog, numbness and tingling, and mood changes. It is one of the more satisfying findings in this whole workup because it is correctable.

The catch is that serum B12 alone is an imperfect test. There is a recognized borderline zone in which a serum B12 result cannot settle the question on its own, and in that zone, adding methylmalonic acid (MMA) or homocysteine gives you a functional readout of whether your cells are actually able to use the B12 that is circulating (PMID 24942828). This is the same logic that shows up repeatedly in this guide: a number inside the reference range is not automatically a number doing its job.

What to ask for: serum B12, serum folate, and, if your B12 lands in a borderline range or your symptoms are strongly suggestive, methylmalonic acid and homocysteine.

A medication note, and please read the second half of it. Metformin use in type 2 diabetes is associated with lower vitamin B12 levels in a systematic review and meta-analysis (PMID 27130885). If you take metformin, or any other medication, and you are concerned about a lab value, the action is to bring it to your prescriber. It is never to stop or change the medication yourself. Monitoring and supplementation are decisions that belong to the person who prescribed the drug.

Section 2: The Rule-Out Tier, Part Two (Inflammatory Screens, Celiac, Sleep)

Inflammatory and Autoimmune Screens

This is the part of the workup where more testing can actively make things worse, so it is worth understanding before you ask for anything.

Fibromyalgia is not an inflammatory arthritis. It does not damage joints and it does not typically produce the inflammatory markers that rheumatoid arthritis or lupus produce. But those conditions can look similar from the outside early on, and they can also occur alongside fibromyalgia, so a screen is reasonable when the clinical picture warrants it.

Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are the general inflammation markers. In fibromyalgia they are usually normal or near normal. A clearly and persistently elevated ESR or CRP is a signal that something else is going on and deserves investigation. This matters especially if you are over 50 with new shoulder and hip girdle pain and stiffness, because polymyalgia rheumatica is a distinct condition whose classification criteria are built around exactly that presentation plus elevated inflammatory markers, and it responds to entirely different treatment (PMID 22388996).

Rheumatoid factor (RF), anti-CCP antibodies, and antinuclear antibody (ANA) are the more specific autoimmune screens, and they come with a real warning. A review of consecutive rheumatology referrals found that at least half of the ANA, RF, ENA, and anti-dsDNA testing performed before referral was ordered inappropriately, with 20 percent of ANA testing done with no clinical indication at all, and noted that RF and ANA have low positive predictive value in general practice (PMID 32522598). Translated: if you order these tests without a clinical reason, you will collect positive results that mean nothing, and those results generate anxiety, repeat testing, and referrals that lead nowhere.

So how should you use this? Ask your physician whether your specific picture warrants autoimmune screening, and accept a well-reasoned no. If your exam shows no synovitis, no rash, no dry eyes and mouth, no Raynaud's, and normal inflammatory markers, a shotgun autoimmune panel is more likely to confuse your chart than clarify it. If you do have those features, the screening is appropriate and you should push for it.

And if a screen comes back positive: that finding belongs in front of a physician, ideally a rheumatologist, before you draw any conclusions. A low-titer positive ANA is common in the healthy population. It is not a diagnosis.

Celiac Disease and Gluten

Celiac disease can present with musculoskeletal and other symptoms outside the gut, which is part of why it is under-recognized (PMID 31513026). Given how much overlap exists between celiac symptoms and fibromyalgia symptoms, screening is reasonable, and it is a simple blood test.

What to ask for: tissue transglutaminase IgA (tTG-IgA) together with total serum IgA. The total IgA matters because selective IgA deficiency will make the tTG-IgA falsely negative, and current guidance is to test both together (PMID 36602836).

Critical practical point that trips people up constantly: celiac serology is only valid if you are actively eating gluten. If you have already gone gluten free, the test can come back negative regardless of whether you have celiac disease. Guideline guidance is that patients should be on a gluten-containing diet when tested (PMID 36602836). If you have already stopped gluten and want to be tested properly, that is a conversation to have with your physician before you change anything, because a gluten challenge is a medical decision with real symptom consequences.

What about gluten sensitivity without celiac disease? This is a genuinely open area. A case series described 20 fibromyalgia patients without celiac disease who improved on a gluten-free diet, with all of them showing intraepithelial lymphocytosis without villous atrophy on biopsy, and the authors framed it as support for a hypothesis rather than as proof (PMID 24728027). That is a case series, not a controlled trial, and it should be read as a reason for curiosity and a supervised trial, not as an established mechanism.

Testing here deserves a precise answer, because it usually gets a sloppy one. Standard celiac serology is narrow by design. It is built to answer one question, which is whether you have celiac disease, and it is not built to see reactivity outside that question. Specialty laboratories offer considerably wider panels that measure antibodies against individual wheat and gluten peptides rather than against the whole protein, alongside related targets such as non-gluten wheat proteins and the transglutaminase enzymes. The Wheat Zoomer from Vibrant Wellness and Array 3X from Cyrex Laboratories are the two most commonly used in functional medicine. Those are real measurements of real antibodies, run at a level of detail standard celiac testing does not attempt, and they can surface reactivity that a normal tTG-IgA will never show you.

What is not established is the step from a positive peptide result to a diagnosis. The current expert consensus definition of non-celiac gluten sensitivity, known as the Salerno criteria, is built entirely on symptom response: celiac disease and wheat allergy ruled out first, then symptoms that improve when gluten is withdrawn and return when it is reintroduced, ideally under blinded conditions (PMID 26096570). There is no antibody cutoff anywhere in that definition, because no antibody has been validated as the thing that settles it.

So the honest way to use one of these panels is as information that helps you and your clinician decide whether a careful elimination and reintroduction is worth doing, and as a baseline you can retest later to watch a direction of travel. It is a reason to run the trial, not a substitute for it. A practitioner who tells you a peptide panel diagnoses gluten sensitivity is going further than the evidence goes. A practitioner who tells you nothing can be measured at all is behind where the laboratories actually are. Both statements are wrong, and the space between them is where the useful work happens.

Sleep, Which Is Not a Blood Test but Belongs Here Anyway

Unrefreshing sleep is one of the core features of fibromyalgia, and it is also the core feature of obstructive sleep apnea. A systematic review and meta-analysis found that approximately 21 percent of patients with obstructive sleep apnea-hypopnea syndrome met criteria for fibromyalgia (PMID 38831795).

You cannot sort this out with a blood draw. If you snore, wake unrefreshed, wake gasping, have witnessed apneas, have morning headaches, or fall asleep during the day, ask your physician about a sleep study. Treating sleep apnea will not cure fibromyalgia. But leaving untreated apnea in place while you work on everything else means you are trying to repair a system that never gets to run its overnight repair cycle.

Section 3: The Terrain Tier (Blood Sugar, Inflammation, Stress Axis)

Everything above is about ruling things out. This section is about describing the environment your nervous system is working in. These are not fibromyalgia tests. They are levers.

Blood Sugar and Insulin

Insulin resistance is worth looking at here, with the caveat that this specific literature requires care.

An observational study reported an association between insulin resistance, measured by HbA1c, and central pain in patients with fibromyalgia (PMID 33740353). You should also know that an earlier paper from the same group, titled “Is insulin resistance the cause of fibromyalgia? A preliminary report,” has been **retracted** and is not cited anywhere in this guide. I am telling you that explicitly because you are likely to find that retracted paper circulating in fibromyalgia communities online, presented as evidence that insulin resistance causes fibromyalgia. It is not usable evidence. The association is worth investigating. The causal claim is not supported.

What to ask for: fasting glucose, HbA1c, and fasting insulin. Fasting insulin is the one that usually gets left off and is often the most informative of the three, because insulin can climb for years while fasting glucose still reads normal. HOMA-IR is not a separate test; it is a calculation your clinician can run from your fasting glucose and fasting insulin.

What a result actually tells you. Rising fasting insulin with normal glucose describes a metabolic pattern worth addressing through nutrition, movement, and sleep. It does not diagnose fibromyalgia, it does not explain your pain by itself, and it is not a reason to panic. It is a modifiable finding, which is exactly what makes it worth measuring.

Inflammatory Markers, Read Carefully

Fibromyalgia is not classically an inflammatory disease, but there is evidence of low-grade inflammatory signal in at least a subset of patients.

A study of 105 fibromyalgia patients and 61 healthy controls found high-sensitivity CRP levels only marginally higher in the fibromyalgia group overall, but with a significantly higher proportion of abnormal values, and the hsCRP values correlated with body mass index, ESR, IL-6, and IL-8. Notably, ESR, IL-6, and IL-8 did not themselves differ between the groups (PMID 23124693). A more recent systematic review and meta-analysis of 18 observational studies found elevated hsCRP in fibromyalgia relative to controls as the one biomarker reaching statistical significance in quantitative synthesis, while hair cortisol, IL-6, and interferon gamma showed no significant between-group differences, and the authors explicitly noted moderate methodological quality, high heterogeneity, and limited direct clinical applicability (PMID 42477737). At the time of writing this reference is published online ahead of print.

How to use that. An elevated hs-CRP in your results is worth knowing and worth addressing, because it points toward inflammatory load that is often modifiable. It is not a fibromyalgia marker, it does not track your pain reliably, and it correlates strongly with body mass index, which means it is telling you about your metabolic terrain as much as anything else.

The Stress Axis (HPA Axis and Cortisol)

The hypothalamic-pituitary-adrenal axis is the system that governs your cortisol rhythm, your stress response, and a good deal of your energy and sleep regulation. Disturbances in this axis have been described in fibromyalgia since at least the mid 1990s (PMID 7980669).

Here is the part that most functional medicine marketing leaves out, and that you deserve to have: the relationship between cortisol findings and fibromyalgia pain is not clean. A study of salivary cortisol release and HPA feedback sensitivity in fibromyalgia found that the alterations tracked with depression rather than with pain (PMID 20627822). And the meta-analysis referenced above found no significant between-group difference in hair cortisol (PMID 42477737).

What that means for you practically. Cortisol testing, whether salivary, urinary, or serum, can be genuinely informative about your stress physiology, your sleep-wake rhythm, and your energy pattern across a day. It is a reasonable thing to run when fatigue, sleep disruption, and stress load dominate your picture. What it is not is a fibromyalgia test, and a "flat cortisol curve" result is not a diagnosis of anything. Treat it as one description of your terrain, interpreted next to everything else, by a clinician.

Also note: true adrenal insufficiency is a serious medical condition diagnosed with specific testing by a physician, and it is a completely different thing from the cortisol rhythm patterns discussed here. If you have unexplained weight loss, low blood pressure, salt craving, darkening of the skin, or episodes of collapse, that needs a physician now.

Magnesium and Other Nutrients

Magnesium is one of the most frequently discussed nutrients in fibromyalgia, and the honest state of the evidence is thinner than the internet suggests. An observational cross-sectional study reported an association between serum magnesium levels and both sleep quality and disease severity in fibromyalgia (PMID 40696633). That is a single cross-sectional study, and association is not causation.

There is also a real technical problem worth knowing: serum magnesium reflects only a small fraction of total body magnesium and can look normal when tissue stores are not. That is a recognized limitation of the test rather than a reason to distrust the lab.

Practical takeaway. Serum magnesium is inexpensive and can be added to a panel. Read a low result as meaningful and a normal result as inconclusive rather than reassuring. Any decision about supplementation belongs with your clinician, particularly if you have kidney disease, where magnesium supplementation carries real risk.

Small Fiber Neuropathy

A meaningful subset of people diagnosed with fibromyalgia have evidence of small fiber neuropathy, which can be assessed by measuring intraepidermal nerve fiber density on a skin punch biopsy (PMID 33802768). This matters if your symptoms include prominent burning, electric, or shooting pain, or numbness and tingling, on top of the widespread aching.

This is not a routine test and it is not something to request casually. It is a conversation to have with a neurologist or a physician who works with nerve symptoms, and the reason to have it is that a small fiber neuropathy finding can change the workup, pointing toward causes such as diabetes, B12 deficiency, or autoimmune conditions that have their own specific management.

Section 4: “In Range” Versus “Optimal,” Explained Honestly

This is the section where you will hear two credible people say different things, so let us be precise about what each one is actually claiming.

What a laboratory reference range is. It is a statistical description of results from a reference population, usually constructed so that roughly the middle 95 percent of that population falls inside it. It is a description of what is common. It was never designed to describe what is ideal, and it is not the same thing as a treatment threshold.

What “optimal range” means in functional medicine. Many functional medicine practitioners, myself included, work with narrower target windows than the laboratory reference range for certain markers, on the reasoning that being at the bottom edge of a broad population range is not the same as being where a given system functions best, and that catching a drift early is better than waiting for it to cross a diagnostic line.

Here is the honest part. For most markers, those narrower windows are a clinical framework, not a validated diagnostic threshold. They come from clinical experience, from physiological reasoning, and from patterns practitioners observe over many patients. They are not, for the most part, numbers that a randomized trial has established as the point where symptoms improve. Anyone presenting a functional optimal range as settled science is overstating it, and you should be skeptical of a guide that does not tell you that.

Where numbers in this guide do come from published sources, I have said so explicitly:

- The vitamin D thresholds of below 20 ng/mL for deficiency and 21 to 29 ng/mL for insufficiency come from a 2011 Endocrine Society guideline (PMID 21646368), and the same body’s 2024 guideline moved away from routine testing and specific general-population targets (PMID 38828931).
- The 32 to 48 ng/mL vitamin D target used in the fibromyalgia randomized trial described earlier is that trial’s protocol target, in 30 women, not a general recommendation (PMID 24438771).
- The ferritin figure of below 50 ng/mL comes from a single case-control study’s finding of increased odds of fibromyalgia at that level (PMID 20087382), and it sits alongside a study that found no iron differences at all (PMID 22095117) and a systematic review indicating that ferritin cutoffs for deficiency are themselves contested (PMID 34028001).

The functional targets I use. Rather than leave you guessing, here are the actual target windows from my blood chemistry framework for the markers this guide covers. Read the caution above first, because it applies to every number in this table: these are clinical targets drawn from functional medicine practice, not validated diagnostic thresholds, and being outside one of them is a question to raise, never a diagnosis to make.

Marker	Standard lab range is roughly	Root Health functional target
TSH	0.4 to 4.0 mIU/L	1.0 to 2.5 mIU/L
Free T4	0.7 to 1.7 ng/dL	0.9 to 1.4 ng/dL
Free T3	2.0 to 4.2 pg/mL	2.8 to 3.6 pg/mL
Reverse T3	up to 24 ng/dL	up to 15 ng/dL
TPO antibodies	up to 35 IU/mL	up to 9 IU/mL
Thyroglobulin antibodies	up to 20 IU/mL	up to 1 IU/mL
Ferritin	25 to 300 ng/mL	50 to 150 ng/mL
Transferrin saturation	15 to 45 percent	20 to 40 percent
Vitamin D (25-OH)	20 to 100 ng/mL	40 to 80 ng/mL
Vitamin B12	200 to 1100 pg/mL	500 to 900 pg/mL
Folate (RBC)	400 to 2000 ng/mL	600 to 1500 ng/mL
Methylmalonic acid	up to 243 nmol/L	up to 162 nmol/L
Homocysteine	up to 10 umol/L	up to 7.0 umol/L
hs-CRP	up to 1.0 mg/L	up to 0.5 mg/L
ESR	up to 25 mm/hr	up to 15 mm/hr
Fasting glucose	70 to 99 mg/dL	75 to 86 mg/dL
HbA1c	4.0 to 5.6 percent	4.6 to 5.3 percent
Fasting insulin	2.0 to 10.0 uIU/mL	2.6 to 5.0 uIU/mL
HOMA-IR	up to 2.5	0.5 to 1.4
Magnesium (serum)	1.5 to 2.7 mg/dL	2.0 to 2.5 mg/dL
Magnesium (RBC)	4.0 to 7.2 mg/dL	5.2 to 6.8 mg/dL
Cortisol (morning)	6.0 to 25.0 ug/dL	12.0 to 22.0 ug/dL

Marker	Standard lab range is roughly	Root Health functional target
DHEA-S	75 to 450 ug/dL	125 to 350 ug/dL

A few notes on using this table well. Reference ranges printed on your own report may differ from the middle column, because each laboratory sets its own from its own population, so always compare against the range on your actual result. Hemoglobin, hematocrit, and several hormone targets vary by sex and by age, so those are left to your clinician rather than reduced to one line here. And a result sitting just outside a functional target while comfortably inside the standard range is common, is not an emergency, and is exactly the kind of finding that deserves a conversation rather than a supplement purchase.

How to use the functional lens well. Look at where your result sits inside the range, not just whether it is inside. Look at the direction of travel across multiple draws over time, which is far more informative than any single value. Read markers in groups rather than one at a time. And treat a "low normal" result as a question worth asking rather than as an answer. That is the useful core of the functional approach, and none of it requires believing an unvalidated cutoff.

Section 5: Reading Your Results as a Pattern

Individual values are noisy. Patterns are where the information lives. Here are combinations worth noticing, all of which are conversation starters with your clinician rather than conclusions.

Everything normal. This is the most common result and it is genuinely good news. It means the correctable mimics have been checked. From here, the work moves to the terrain tier and to the parts of care that do not depend on labs at all, including sleep, graded movement, nervous system regulation, and nutrition.

Low or low-normal ferritin plus poor sleep plus restless legs. Iron status and sleep quality are worth looking at together rather than separately. This is a correctable-in-principle combination and worth raising specifically.

Positive thyroid antibodies with a normal TSH. Autoimmune thyroid activity without current gland failure. Not a fibromyalgia explanation, but a real finding that warrants monitoring and physician follow-up.

Elevated hs-CRP with a raised body mass index. The published correlation between these two is strong (PMID 23124693), which points the conversation toward metabolic and inflammatory terrain rather than toward an autoimmune diagnosis.

Elevated ESR or CRP that stays elevated. This is the pattern that should push hardest toward further physician workup, because persistent inflammatory marker elevation is not typical of fibromyalgia alone and points toward something additional going on.

Rising fasting insulin with a normal fasting glucose. A metabolic pattern that is modifiable, worth addressing, and not by itself an explanation for widespread pain.

Borderline serum B12 with symptoms that fit. The situation where methylmalonic acid or homocysteine earns its place (PMID 24942828).

A positive autoimmune screen of any kind. Stop interpreting and get a rheumatology opinion. This is the one pattern in the list where self-interpretation carries real cost.

Section 6: The Conversation With Your Doctor

Most of the tests in this guide are ordinary, inexpensive, and orderable by any primary care provider. You do not need a specialist and you do not need a functional medicine practice to get the rule-out tier drawn. What you often do need is to ask specifically, because the default panel is narrower than what is described here.

How to open the conversation

Something like this works well, and it works because it is true:

> *"I understand fibromyalgia is diagnosed clinically and that there is no lab test that confirms it. What I want to do is make sure we have ruled out the treatable conditions that can look like this or make it worse. Can we check thyroid function including antibodies, iron status including ferritin, vitamin D, B12, celiac screening, and basic inflammatory markers?"*

That framing does three things. It shows you understand the diagnosis rather than shopping for a different one. It names rule-out, which is a goal your physician already shares. And it asks for specific tests instead of asking for "everything."

A request list you can bring

Tier one, the rule-out round. Complete blood count with differential. Comprehensive metabolic panel. TSH, free T4, free T3, TPO antibodies, thyroglobulin antibodies. Ferritin, serum iron, TIBC, transferrin saturation. 25-hydroxyvitamin D. Vitamin B12 and folate. Tissue transglutaminase IgA with total IgA. ESR and hs-CRP. Creatine kinase if you have prominent muscle weakness as opposed to pain.

Tier two, added when the picture warrants it. Methylmalonic acid and homocysteine if B12 is borderline. Fasting insulin alongside fasting glucose and HbA1c. Magnesium. Autoimmune screening (ANA, RF, anti-CCP) only if there are clinical features that justify it, for the reasons in Section 2. A sleep study if any sleep apnea features are present. Cortisol rhythm assessment if fatigue and stress physiology dominate. Referral for small fiber neuropathy evaluation if burning, electric, or numb-and-tingling symptoms are prominent.

If your request is declined

Ask why, and listen to the answer, because sometimes it is a good one. "Your inflammatory markers are normal and you have no synovitis, so an ANA is more likely to mislead us than help" is sound medicine, not dismissal. "There is no point testing anything because it is just fibromyalgia" is a different kind of answer, and it is a reasonable trigger to seek a second opinion.

You can also ask your physician to document the request and the reasoning in your chart. That is a normal request, it costs nothing, and it tends to make the conversation more precise.

Practical notes on getting drawn

Fast for 8 to 12 hours if fasting glucose, fasting insulin, or a lipid panel is included, and drink water. If you take biotin, including in a hair, skin, and nail supplement, tell the lab and your physician, because high-dose biotin can interfere with several immunoassays including thyroid tests. Ask for a copy of your own results, in numbers, not just a “normal” phone call. You are entitled to them and you cannot track a trend you never saw.

Section 7: What These Labs Cannot Tell You

Being clear about the limits is part of respecting your time and your money.

They cannot confirm fibromyalgia. Nothing in this guide does that, and any test marketed as a fibromyalgia diagnostic should be approached with real skepticism.

They cannot measure your pain. There is no lab value that tracks how much you hurt. Your reported experience is the primary data on that, and it is legitimate data.

They cannot rank your root causes for you. Several findings often show up at once. Deciding which one to address first, and in what order, is clinical judgment applied to your specific history. That is the work a good clinician does and it is not something a document can do for you.

They cannot substitute for the interventions that actually have the strongest evidence in fibromyalgia, which are not laboratory findings at all. Graded physical activity, sleep, and nervous system regulation carry more weight in the outcome literature than any marker in this guide, and no amount of testing replaces them.

Normal labs do not mean nothing is wrong. Say that one out loud a few times if you need to. It is the single most common misreading in this entire condition, and it has kept a lot of people from getting care they needed.

Section 8: When to Seek Clinical Review

Do not wait on lab results, and do not work through this guide first, if you have any of the following:

- Fever, drenching night sweats, or unexplained weight loss alongside your pain
- Visible joint swelling, warmth, or redness, particularly with morning stiffness lasting more than an hour
- New or worsening muscle weakness, as distinct from pain or fatigue
- New onset of widespread pain and girdle stiffness after age 50, particularly with new headache, jaw pain when chewing, scalp tenderness, or any visual change, which requires urgent evaluation
- Progressive numbness, foot drop, loss of balance, or falls
- Any bladder or bowel control change, or numbness in the saddle region, which is an emergency
- Thoughts of harming yourself, which warrants immediate help through your physician, a crisis line, or emergency services

Fibromyalgia is a real diagnosis, and it is also a diagnosis that can sit on top of something else. Getting a new or changing symptom evaluated is not being difficult. It is good self-advocacy.

Your Next Step

Take this guide, or just the request list in Section 6, to your next appointment. Get the rule-out round drawn. Ask for your results in numbers.

Then bring those numbers to someone who will read them as a pattern rather than as a pass or fail. If you want that conversation with me, a Root Discovery Session is complimentary and exists for exactly this purpose:

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

Clinical care is limited to people located in North Carolina, where I hold my license. That is a licensing boundary, and it is worth knowing that it is narrower than it sounds.

If you live outside North Carolina, education is not restricted by state lines, and neither is your own record keeping. Take the request list in Section 6 to your own clinician, and start recording your results and symptoms over time. A single draw is a snapshot. The direction of travel across several draws is where the information actually lives, and no one will keep that record for you:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

And if you want a second set of eyes on the pattern you are seeing, write to me and say where you are located. Education, second opinions on how to read a trend, and help deciding what is worth asking your own clinician for are all things I can offer wherever you live. I will tell you plainly what I can do from here and what belongs with someone licensed in your state:

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

References

Every paper below was pulled and read against its original record, and checked to be sure it had not been retracted. One paper that came up in the research had been, and instead of quietly leaving it out I named it in the blood sugar section, because it still circulates in fibromyalgia groups as proof of something it never proved.

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